

Bhushan Ladgaonkar

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PROFESSIONAL SUMMARY

Security Researcher and Machine Learning Engineer specializing in **Adversarial AI** and **AI Security**. Published first-author researcher in adversarial robustness; proven experience engineering ML-based Network Intrusion Detection Systems for enterprise infrastructure and building hardware-optimized local LLM inference systems on Apple Silicon. Seeking to leverage expertise in adversarial robustness, simulation, and security architectures.

EDUCATION

Fr. Conceicao Rodrigues College of Engineering
Bachelor of Technology (B.Tech) in Computer Engineering

Mumbai, India
Expected May 2028

EXPERIENCE

Oil and Natural Gas Corporation (ONGC)
Summer Intern – Database and Security Group

Mumbai, India
May 2025 – June 2025

- Developed a prototype **Network Intrusion Detection System (NIDS)** on the UNSW-NB15 dataset (2M+ records) to identify zero-day threats, addressing limitations of traditional signature-based detection.
- Implemented **Random Forest** and **XGBoost** classifiers with **SMOTE**-based class imbalance handling and advanced feature engineering, reducing false positive rates while sustaining high detection sensitivity.
- Collaborated with the Database Group to document threat detection methodologies, translating raw data findings into actionable security intelligence.

PUBLICATIONS

Benchmarking Resilience: Asymmetric Latent Purification Inspired by Generative Diffusion Bottlenecks *Published: May 2026*

First & Corresponding Author · International Journal of Research and Innovation in Applied Science (IJRIAS) · doi.org/10.51584/IJRIAS.2026.110400185

- Proposed the **Asymmetric Latent Purifier (ALP)**: a zero-shot, single-pass structural defense that weaponizes a stochastic 64-channel latent bottleneck to shatter PGD adversarial gradients before they reach the classifier – eliminating the need for full generative models.
- Recovered CIFAR-10 classification accuracy from **0.00% → 32.06%** (SD = 1.94%) under iterative **PGD-7** attack, maintaining **25.68 dB PSNR** structural fidelity.
- Optimized inference to **1.25 ms/sample** on Apple Silicon (MLX) – **100×–1000× faster** than iterative diffusion purifiers – trained exclusively on clean images with zero adversarial examples.

TECHNICAL PROJECTS

Gemma-M4-Agent: Local LLM Inference System | *Python, MLX, FastAPI, ChromaDB, React* Apr 2026 – Ongoing

- Built a full-stack local LLM agent around **Google Gemma** on Apple Silicon M4 via **MLX framework** – direct hardware-to-model ownership without wrappers such as Ollama or LM Studio; 8-bit quantized model footprint of **8.5 GB** on Unified Memory.
- Engineered a **Hybrid Memory Architecture** decoupling SSD-resident model weights from USB-persistent **ChromaDB** vector store (384-dim embeddings), enabling zero-copy GPU inference with persistent long-term memory across reboots.
- Implemented **dual-gate stop-token defense**: silicon-level halt via MLX kernel token IDs and software-level control-sequence buffer matching, achieving **100% stop-token accuracy** and preventing hallucinated output beyond semantic boundaries.

TISD-Edu-AI: Local-First RAG Tutor | *Python, MLX, ChromaDB, FastAPI, LLMs*

Apr 2026

- Engineered a privacy-focused on-device **RAG** pipeline using **Microsoft Phi-3 (3.8B)** and **ChromaDB**, achieving **41.42 tokens/sec** at 1.72 s latency with a 9.99 GB footprint on Apple Silicon via MLX.
- Ingested 4,600+ pages of academic PDFs into 6,400+ semantic chunks; validated with a custom evaluation suite achieving **81.1% semantic accuracy** across a 100-sample stress test.

TECHNICAL SKILLS

Languages: Python, C, SQL, Java

AI and ML: Machine Learning, Reinforcement Learning, RAG, LLMs, MCTS, PyTorch, MLX, ChromaDB, XGBoost

Security: Network Intrusion Detection (NIDS), Adversarial Robustness, OSINT, Reconnaissance, Linux (Kali, Ubuntu)

Developer Tools: Git, FastAPI, React, Node.js, Docker, VS Code, Jupyter Notebooks

CERTIFICATIONS

CS50 Cybersecurity – Harvard University

Comprehensive study of encryption, web security, network protocols, and defensive programming practices.

Offensive Security Operations – Cybrary

Hands-on training in penetration testing methodologies, exploitation techniques, and red team operations.

OSINT (Open-Source Intelligence) – Cybrary

Techniques for gathering, analyzing, and operationalizing publicly available intelligence for threat investigations.

Cyber Kill Chain – Cybrary

Framework for understanding attacker TTPs across all intrusion stages to improve detection and response.

Reconnaissance & Enumeration Basics – Cybrary

Foundational skills in active/passive reconnaissance and target enumeration for security assessments.